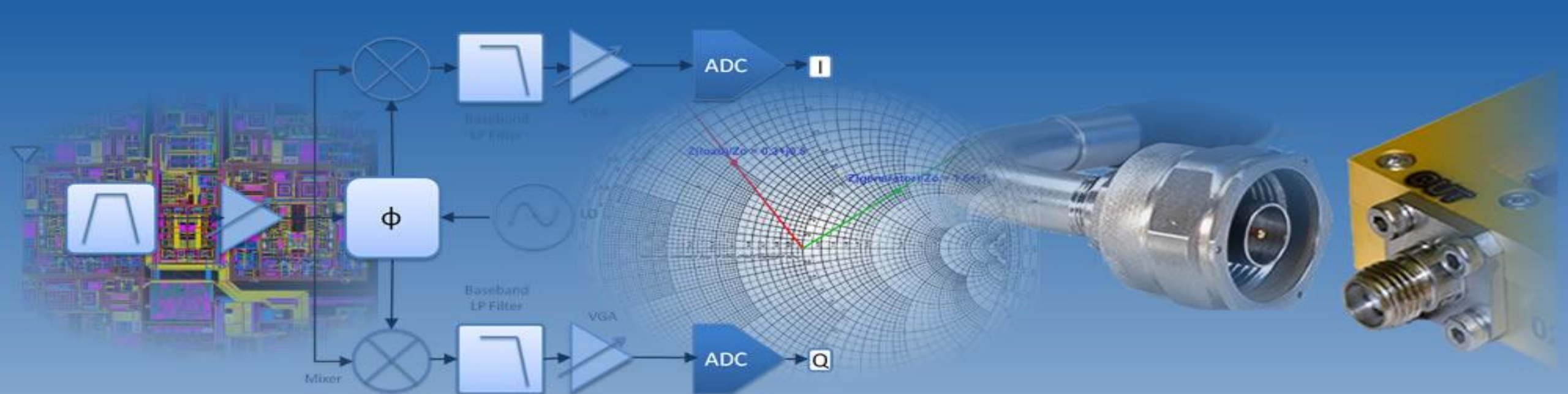




RF & Microwave



Course Structure

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- ▶ **Theory: 80**
 - ▶ **Chapter 1: Introduction** **8 marks**
 - ▶ **Chapter 2: RF and M/W Transmission Lines** **10 marks**
 - ▶ **Chapter 3: RF and M/W Network Theory and Analysis** **8 marks**
 - ▶ **Chapter 4: RF/MW Components and Devices** **10 marks**
 - ▶ **Chapter 5: Microwave Generator** **8 marks**
 - ▶ **Chapter 6: RF Design Practice** **20 marks**
 - ▶ **Chapter 7: Microwave Antennas and Propagation** **8 marks**
 - ▶ **Chapter 8: RF/MW Measurements** **8 marks**
- ▶ **Practical: 25**

Reference Books

1. Microwave Electronics- K.C Gupta, Tata McGraw Hill
2. Microwave Engineering- A.K. Gautam, S.K.Kataria & Sons
3. Microwave Techniques-D.C. Agrawal, Tata Mc Graw-hill
4. Microwave Devices and Circuits-Samuel Y. Liao, PHI 3rd Edition,1994
5. Microwave Engineering-David M. Pozar, 2nd Edition, Newington CT: 1997
6. Engineering Electromagnetic-W.H. Hyatt, McGraw-Hill Book Company
7. Electronic Transmission Technology: Lines, Waves and Antennas-William Sinnema, Prentice Hall

Chapter 1: Introduction

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ER. KOBID KARKEE

RF/Microwave

- ▶ RF is a rate of oscillation in the range around 3KHz to 300MHz, which corresponds to the frequency of radio waves and the alternating currents which carry radio signals.
- ▶ Microwave: is not meant to suggest a wavelength micrometer, have smaller wavelength compared to radio broadcasting.
 - ▶ Frequency Range: 300MHz-300GHz
 - ▶ Wavelength: 1mm – 1m
 - ▶ Includes UHF (0.3-3GHz), SHF (3-30GHz) and EHF (30-300GHz) signals.

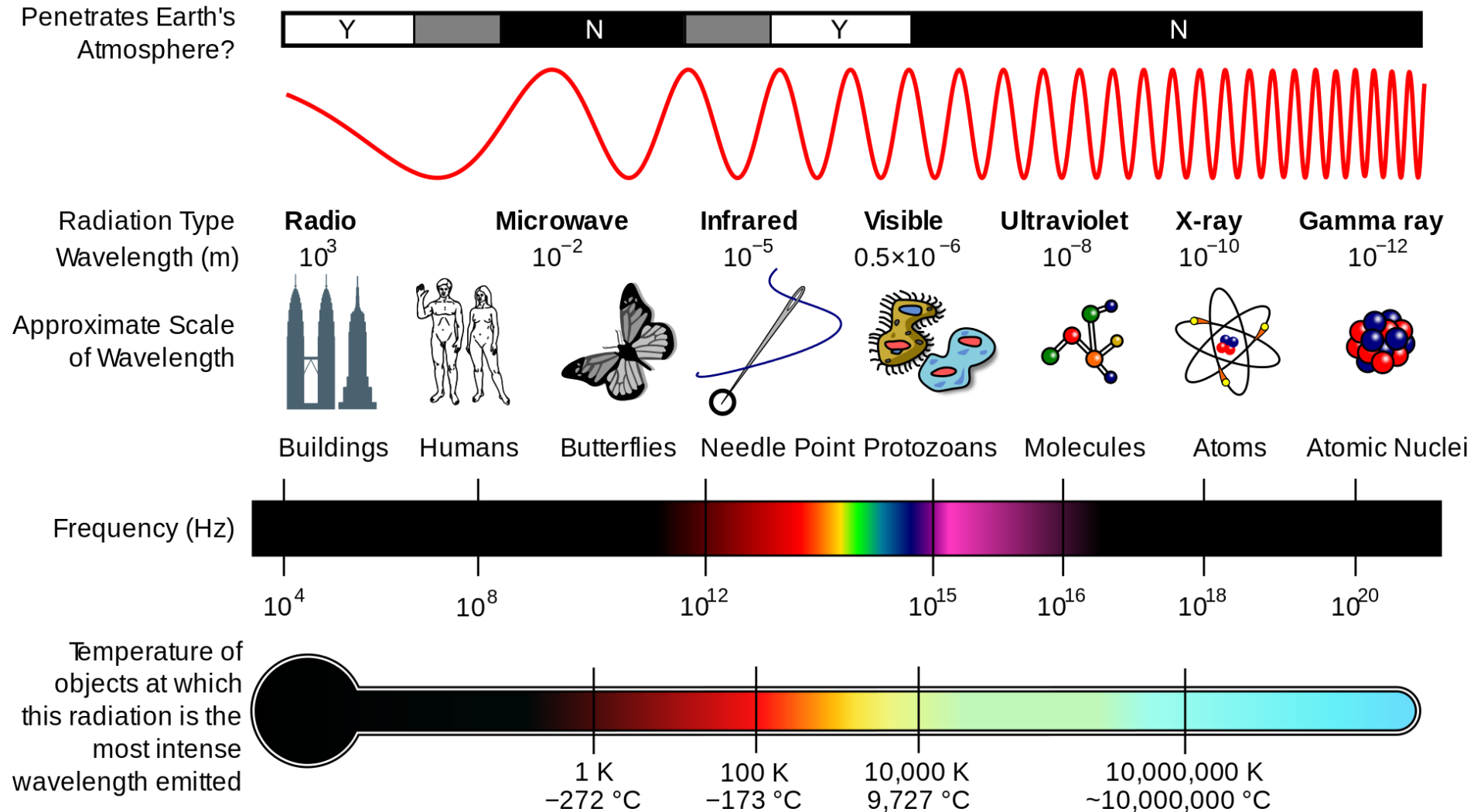
History of Microwave

- ▶ James Clerk Maxwell (1831-1879)
 - ▶ Unified all previous known results, experimental and theoretical on electromagnetic waves in four equations.
 - ▶ Predicted the existence of EM waves.
- ▶ Heinrich Rudolf Hertz (1857-1937)
 - ▶ Experimentally confirmed Maxwell's prediction.
- ▶ Guglielmo Marconi (1874-1937)
 - ▶ Transmitted information on an experimental basis at microwave frequencies.
- ▶ George C. Southworth (1930)
 - ▶ Really carried out Marconi's experiments on a commercial basis.
- ▶ During World War II (1945)
 - ▶ Radar was invented and was exploited for military applications.

Electromagnetic Spectrum

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Electromagnetic Spectrum

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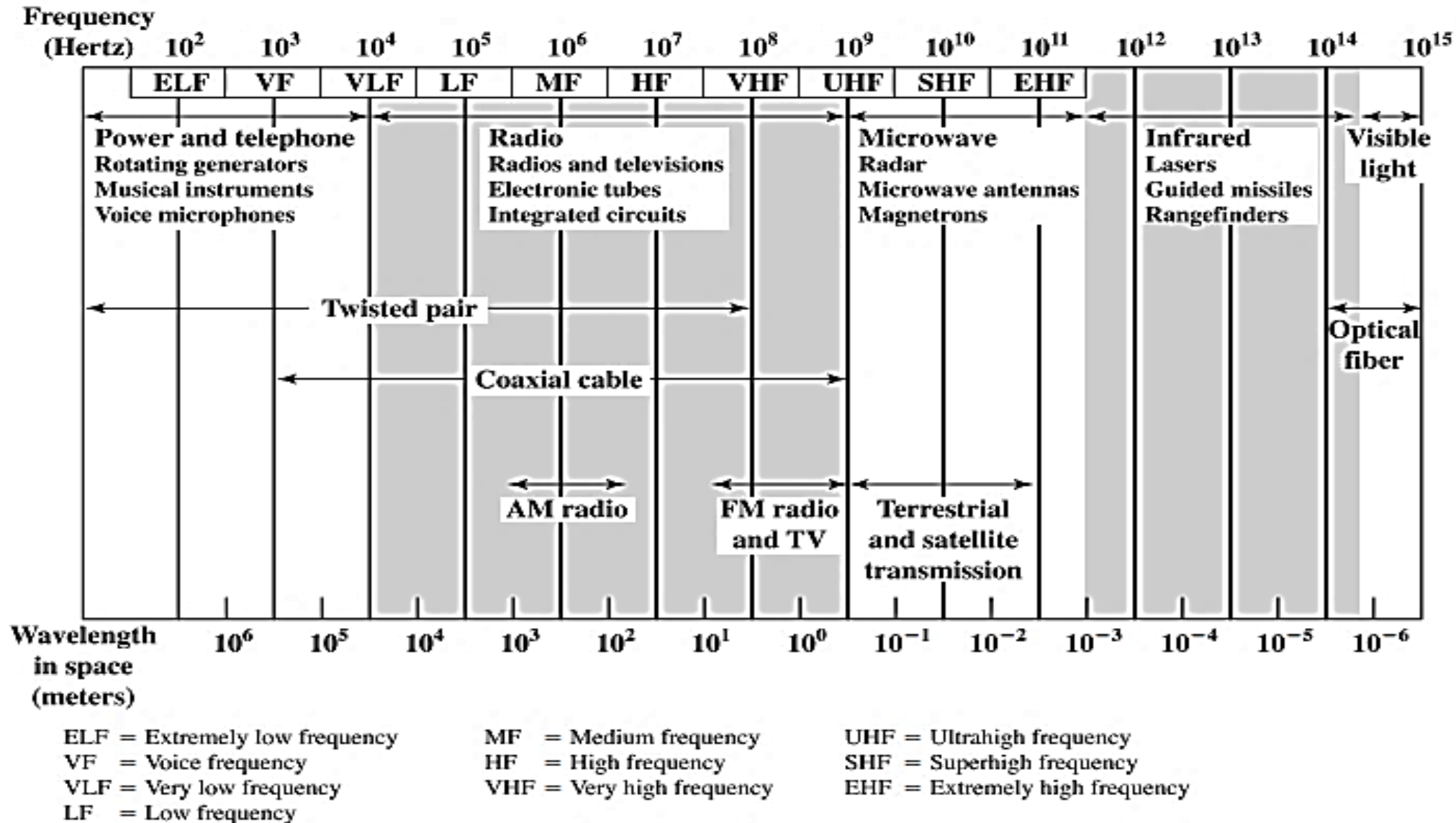
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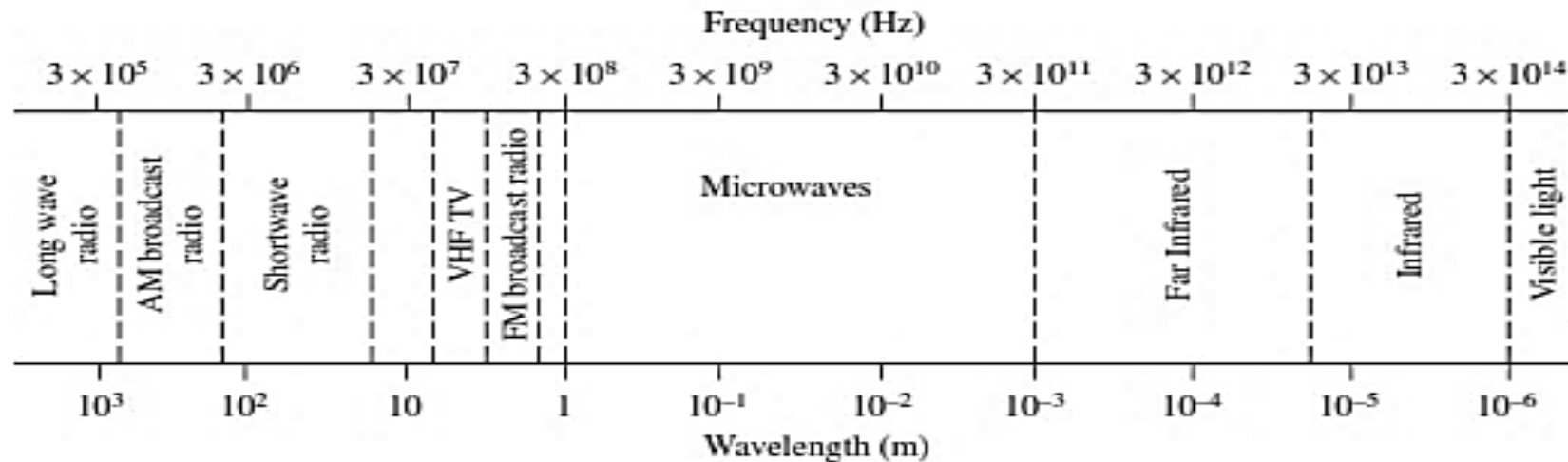
TECHNOLOGIES IN THE ELECTROMAGNETIC SPECTRUM

NONIONIZING RADIATION RADIO FREQUENCIES AND MICROWAVES						IONIZING RADIATION X-RAYS AND GAMMA RAYS		
Extremely Low Frequency	Very Low Frequency	Radio Frequencies	Microwaves	Infrared	Visible Light	Ultra-violet	Soft X-ray	Hard X-ray Gamma Rays
Power lines Hydro lines Computer terminals Electric appliances Telephones Mobile phones Cell phones	Navigation Computer terminals Pagers Mobile phones Cell phones	Radio / CB Television Computer terminals Mobile phones Pagers Cell phones	Radar Computer terminals Microwave ovens Mobile phones (@2.5GHz) Cell phones	Sunlight Computer terminals		Sunlight Computer terminals	Medical X-rays Television Computer terminals	Nuclear fall out
Possible Health Effects Associated With Overexposure Based On Scientific Studies								
Blood disorders Leukemia Cancer Cell growth Embryonic effect	CNS effects Central Nervous System Immune System Cell Membrane Effects	Cataracts Miscarriages Birth defects Blood disorders Embryonic effect CNS effects Central Nervous System	Cataracts Miscarriages Birth defects Genetic damage Birth defects CNS effects Central Nervous System	Cataracts		Skin cancer Cataracts	Cancer Genetic damage Premature aging	

Electromagnetic Spectrum

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Typical Frequencies

AM broadcast band	535–1605 kHz
Short wave radio band	3–30 MHz
FM broadcast band	88–108 MHz
VHF TV (2–4)	54–72 MHz
VHF TV (5–6)	76–88 MHz
UHF TV (7–13)	174–216 MHz
UHF TV (14–83)	470–890 MHz
US cellular telephone	824–849 MHz
	869–894 MHz
European GSM cellular	880–915 MHz
	925–960 MHz
GPS	1575.42 MHz
	1227.60 MHz
Microwave ovens	2.45 GHz
US DBS	11.7–12.5 GHz
US ISM bands	902–928 MHz
	2.400–2.484 GHz
	5.725–5.850 GHz
US UWB radio	3.1–10.6 GHz

Approximate Band Designations

Medium frequency	300 kHz–3 MHz
High frequency (HF)	3 MHz–30 MHz
Very high frequency (VHF)	30 MHz–300 MHz
Ultra high frequency (UHF)	300 MHz–3 GHz
L band	1–2 GHz
S band	2–4 GHz
C band	4–8 GHz
X band	8–12 GHz
Ku band	12–18 GHz
K band	18–26 GHz
Ka band	26–40 GHz
U band	40–60 GHz
V band	50–75 GHz
E band	60–90 GHz
W band	75–110 GHz
F band	90–140 GHz

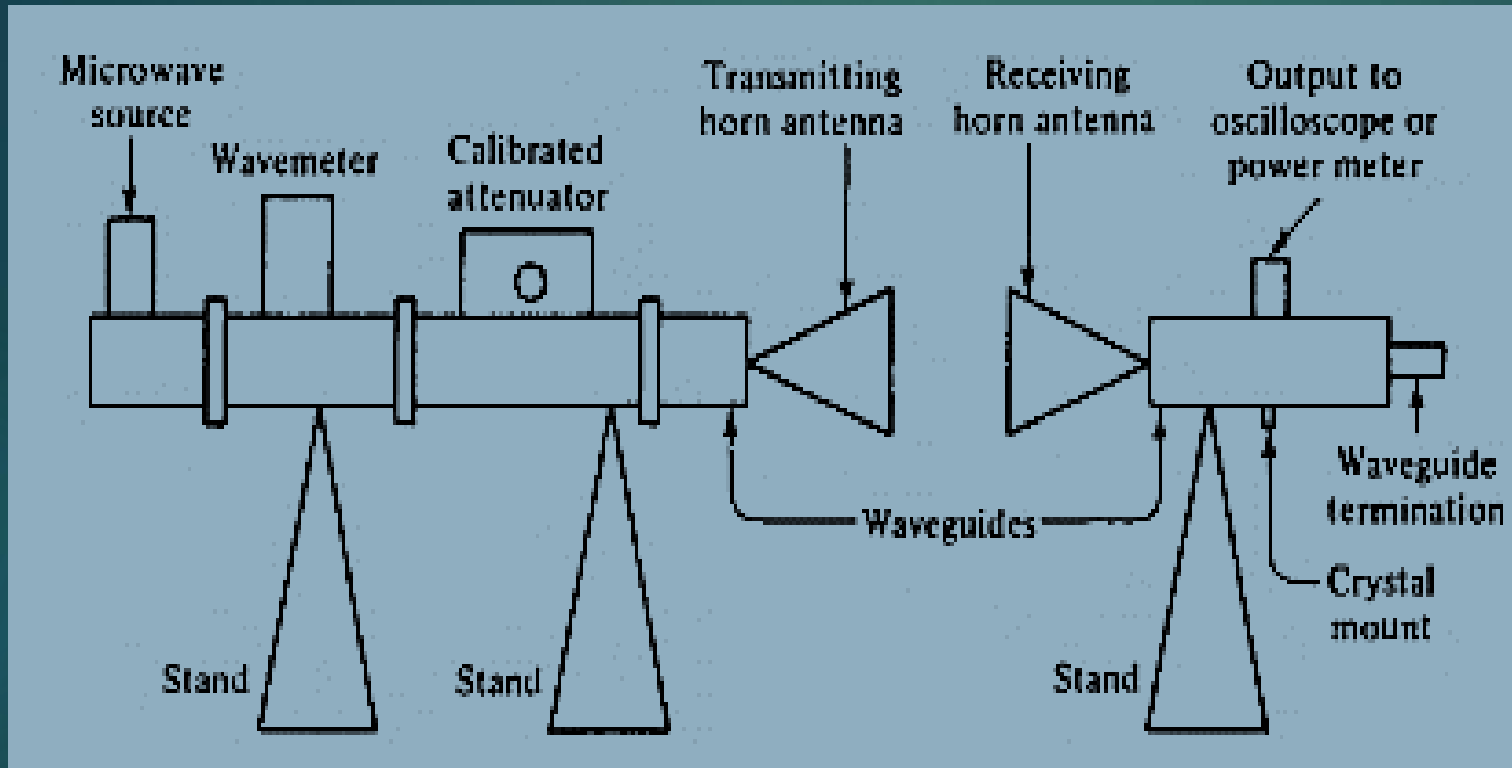
IEEE Microwave Frequency Band

Band	Frequency range	wavelength range [cm]
HF	3-30 MHz	10-100 m
VHF	30-300 MHz	1-10 m
UHF	300-1000 MHz	30-100 cm
L	1-2 GHz	15-30 cm
S	2-4 GHz	7.5-15 cm
C	4-8 GHz	3.75-7.5 cm
X	8-12 GHz	2.5-3.75 cm
Ku	12-18 GHz	16.7-25 mm
K	18-27 GHz	11.1-16.7 mm
Ka	27-40 GHz	7.5-11.1 mm
V	40-75 GHz	4.0-7.5 mm
W	75-110 GHz	2.7-4.0 mm
mm	110-170 GHz	1.8-2.7 mm

Microwave System

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Microwave Sources

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- ▶ High power microwave sources use specialized vacuum tubes to generate microwave.
- ▶ Operate on the principle of ballistic motion of electrons in a vacuum under the influence of controlling electric or magnetic field.
- ▶ Includes magnetron, klystron, travelling wave tube (TWT) and gyrotron.
- ▶ These devices work in the density modulated mode. (work on the basis of clumps of electrons flying ballistically through them, rather than using a continuous stream of electrons).

- ▶ Lower power microwave sources use solid state devices such as the FET, tunnel diodes, Gunn diodes and IMPATT diodes.
- ▶ MASER is a solid state device which amplifies microwave using similar principle to the LASER, which amplifies higher frequency light waves.

Advantages of using higher frequency

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- ▶ Larger instantaneous bandwidth for much information
- ▶ Higher resolution for radar imaging and sensing
- ▶ Reduced dimensions for components
- ▶ Less interference by near by application
- ▶ Higher speed for digital communication, signal processing and transmission
- ▶ Less crowded spectrum
- ▶ Difficulty in jamming (military application)

Disadvantages of using high frequency

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- ▶ More expensive component
- ▶ Higher atmospheric loss
- ▶ Reliance in GaAs technology rather than Si technology
- ▶ Higher component losses, lower output power from active devices
- ▶ Less accurate design tools, less mature tools

Microwave Applications

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► Communication

- Before the advent of fiber-optic transmission, most long-distance telephone calls were carried via networks of microwave radio relay links using FDM.
- Wireless LAN protocols, such as Bluetooth and the IEEE 802.11 specifications, also use microwaves in the 2.4 GHz ISM band.
- Wireless internet access IEEE 802.11a uses microwaves at 3.5-4 GHz range.
- Mobile phone networks, like GSM, use the low-microwave/high-UHF frequencies around 1.8 and 1.9 GHz in the Americas and elsewhere, respectively.

Microwave Applications

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► Communication

- 3G and 4G technologies also use microwave frequencies. (What are the bands???)
- Metropolitan area network (MAN) protocols, such as WiMAX (Worldwide Interoperability for Microwave Access) are based on standards such as IEEE 802.16, designed to operate between 2 to 11 GHz.
- Mobile Broadband Wireless Access (MBWA) protocols based on standards specifications such as IEEE 802.20 operate between 1.6 and 2.3 GHz.

Microwave Applications

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► Communication

- Most satellite communications systems operate in the C, X, Ka, or Ku bands of the microwave spectrum.
- Satellite TV either operates in the C band for the traditional large dish fixed satellite service or Ku band for direct-broadcast satellite.
- Military communications run primarily over X or Ku-band links.

Microwave Applications

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▶ Commercial Applications

- ▶ Commercial implementations are in the 2.3 GHz, 2.5 GHz, 3.5 GHz and 5.8 GHz ranges (ISM band- Industry, Science and Medicine band).

▶ Navigation

- ▶ Global Navigation Satellite Systems (GNSS) broadcast navigational signals in various bands between about 1.2 GHz and 1.6 GHz.
- ▶ These include the Chinese Beidou, the American Global Positioning System (GPS), the European Galileo and the Russian GLONASS.

Microwave Applications

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▶ RADAR

- ▶ Radar uses microwave radiation to detect the range, speed, and other characteristics of remote objects.
- ▶ Weather prediction
- ▶ Geological survey for natural resource exploration.
- ▶ Air traffic control, road traffic surveillance.

Microwave Applications

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- ▶ Radio astronomy

- ▶ Celestial body research.
- ▶ Distance measurement.

- ▶ Heating and Power Application

- ▶ A microwave oven uses microwave radiation at a frequency near 2.45 GHz.
- ▶ Dielectric heating through energy absorption.
- ▶ Microwave heating is used in industrial processes for drying and curing products.

Microwave Applications

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▶ Security

- ▶ RFID based identification system.
- ▶ Motion detectors.

▶ Medical Use

- ▶ Photoacoustic imaging (ultrasound).
- ▶ CT scan, X-rays, MRI like imaging.
- ▶ Thermo-therapy, LASER therapy.
- ▶ Cancer treatment.

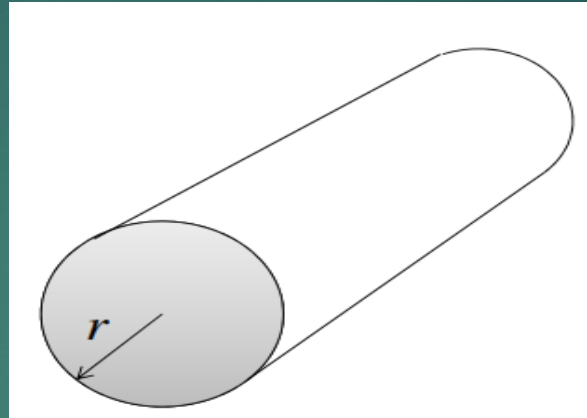
Behavior of circuits at conventional and RF/Microwave bands

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Resistor

$$R_{dc} = \frac{l}{\sigma A}$$

$$R_{dc} = \frac{1}{\sigma \pi r^2} \quad \Omega / m$$

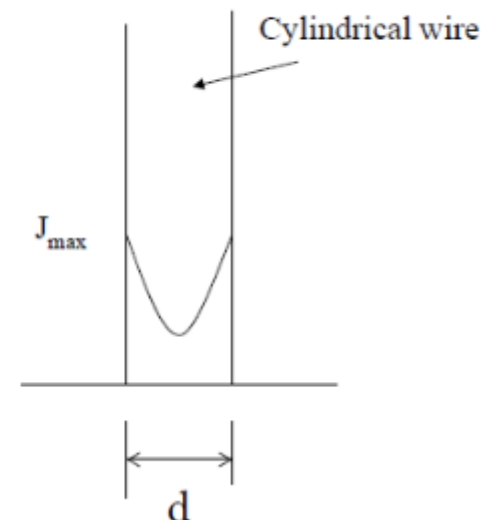


Resistance

AC $\rightarrow$ B-fields $\rightarrow$ Force on charge carriers

Skin depth

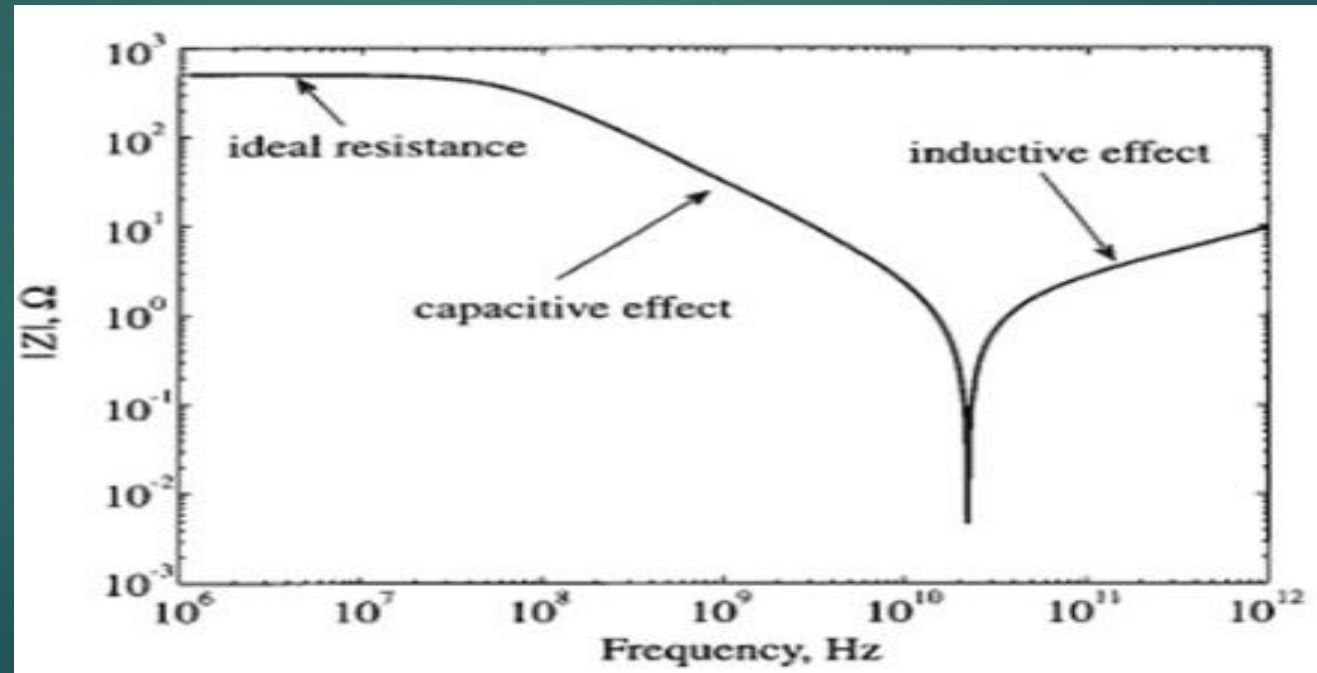
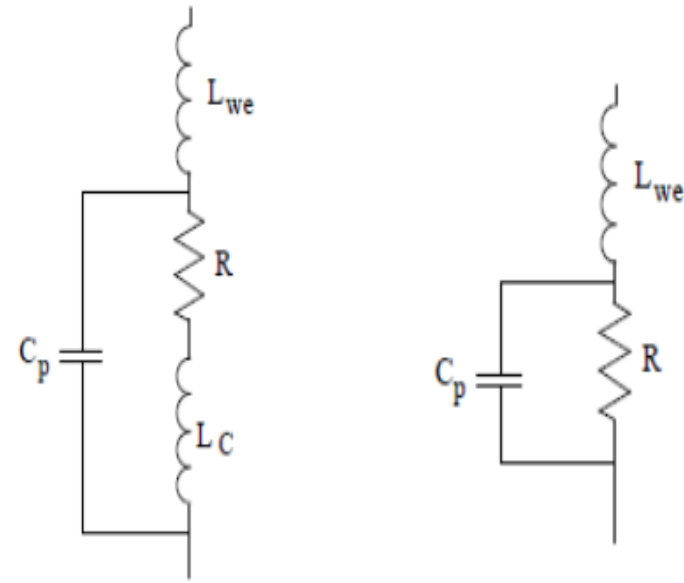
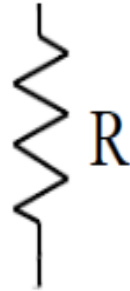
$$\delta = \sqrt{\frac{1}{\pi f \mu \sigma}}$$



$$Z(j\omega) = j\omega L_{we} + (R // \frac{1}{j\omega C_p})$$

$$f_{RC} = \frac{1}{2\pi RC_p}$$

$$f_r = \frac{1}{2\pi\sqrt{L_{we}C_p}}$$

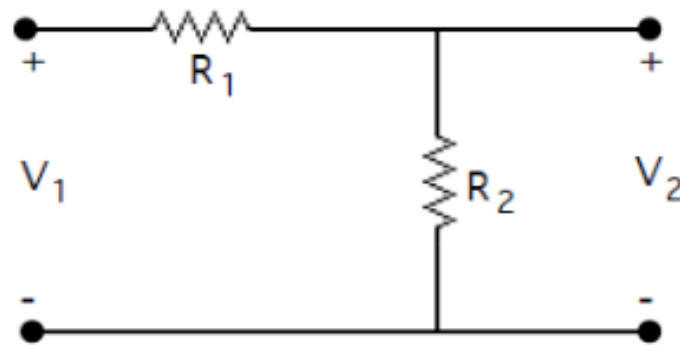


Low value resistor $\rightarrow$ Parasitic inductance as main problem
High value resistor $\rightarrow$ Parasitic capacitance as main problem

Practical tips to avoid this

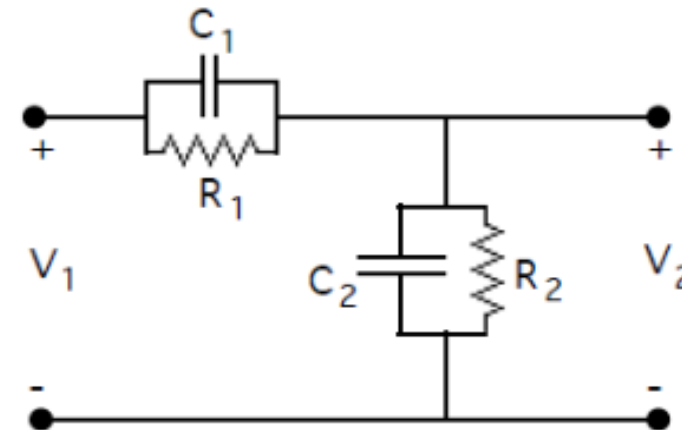
- Use short leads
- low value resistors in series for high resistance
- high value resistors in parallel for low resistance

Practical Example- Voltage divider



Purely resistive

Ideal resistors give a frequency independent transfer function



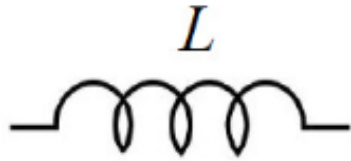
Compensated

C_1 and C_2 provide a path for current at high frequency

Inductor

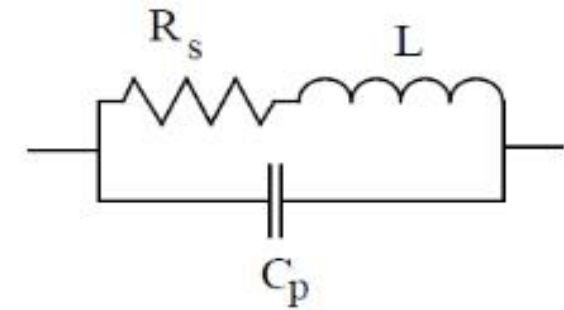
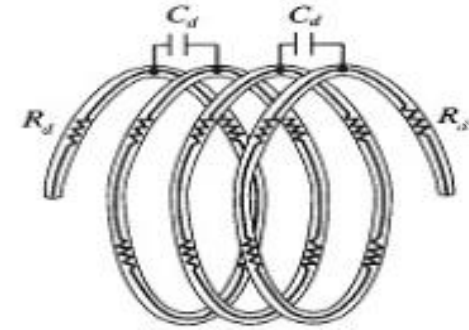
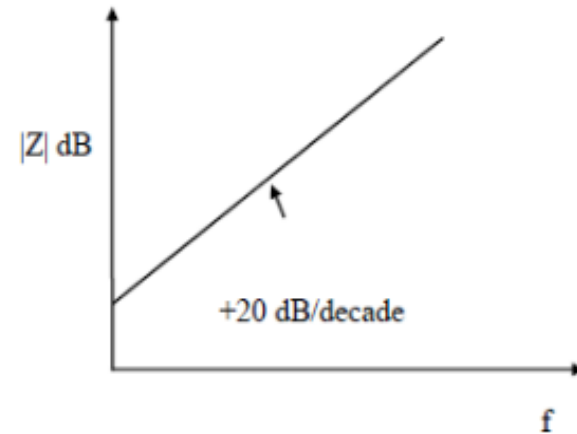
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Ideal

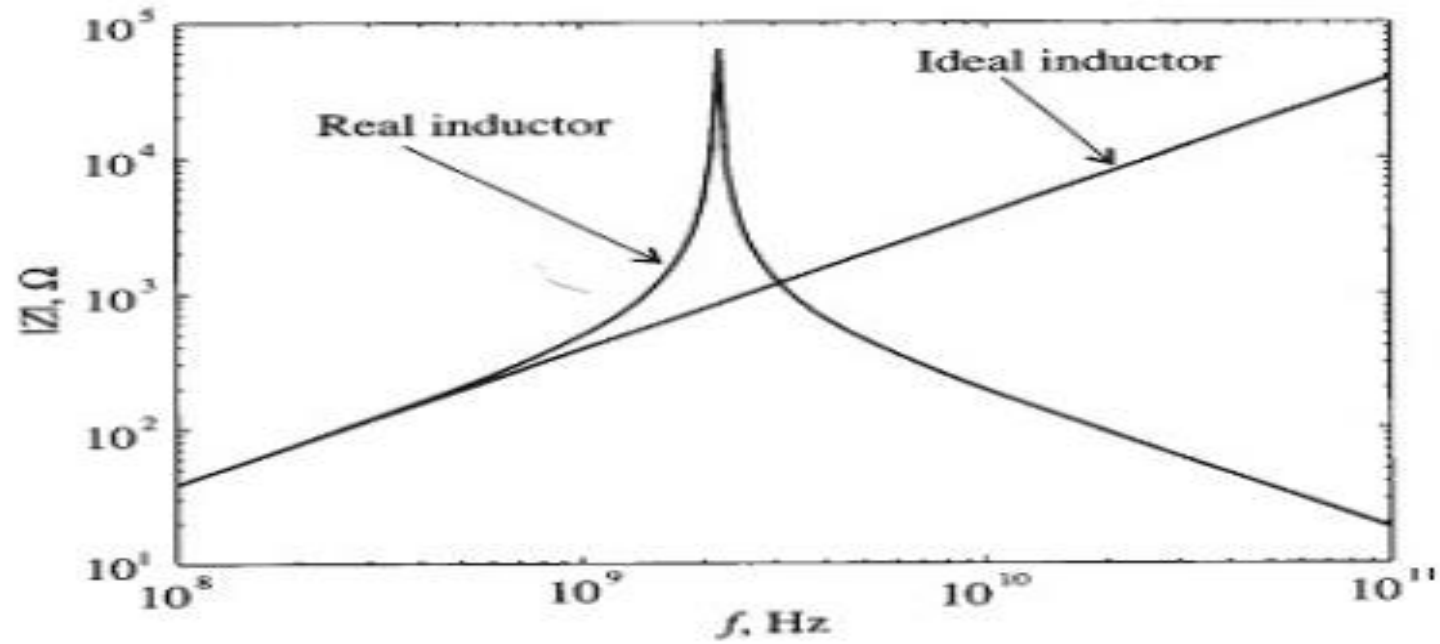


$$Z_L = j\omega L = \omega L \angle +90^\circ$$

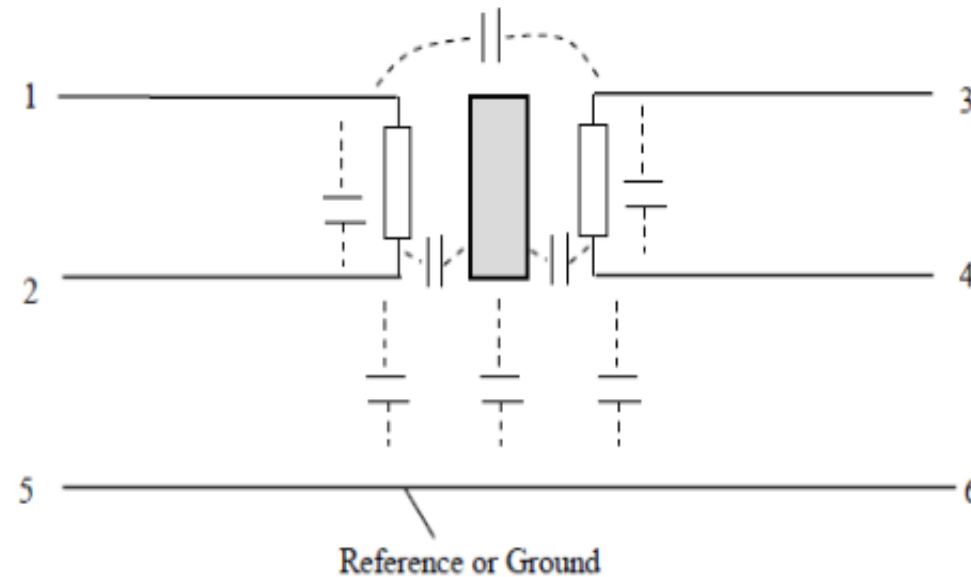
$$|Z_L| = 2\pi L \cdot f$$



$$Z(j\omega) = (R_s + j\omega L) // \frac{1}{j\omega C} \quad f_r = \frac{1}{2\pi\sqrt{LC}}$$



Parasitic capacitances cause the transfer of unwanted signals.

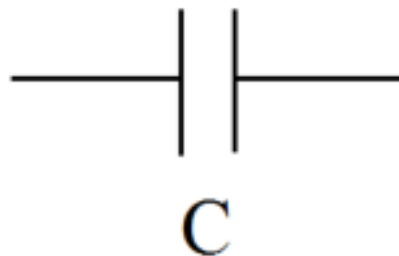


Capacitors

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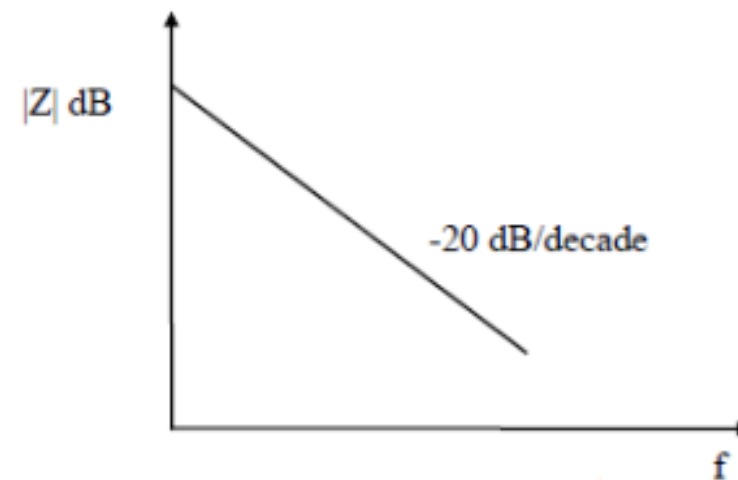
Capacitors

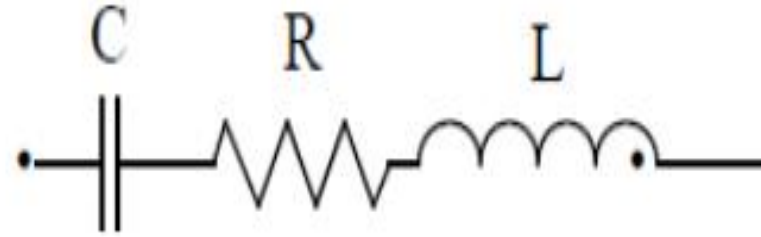
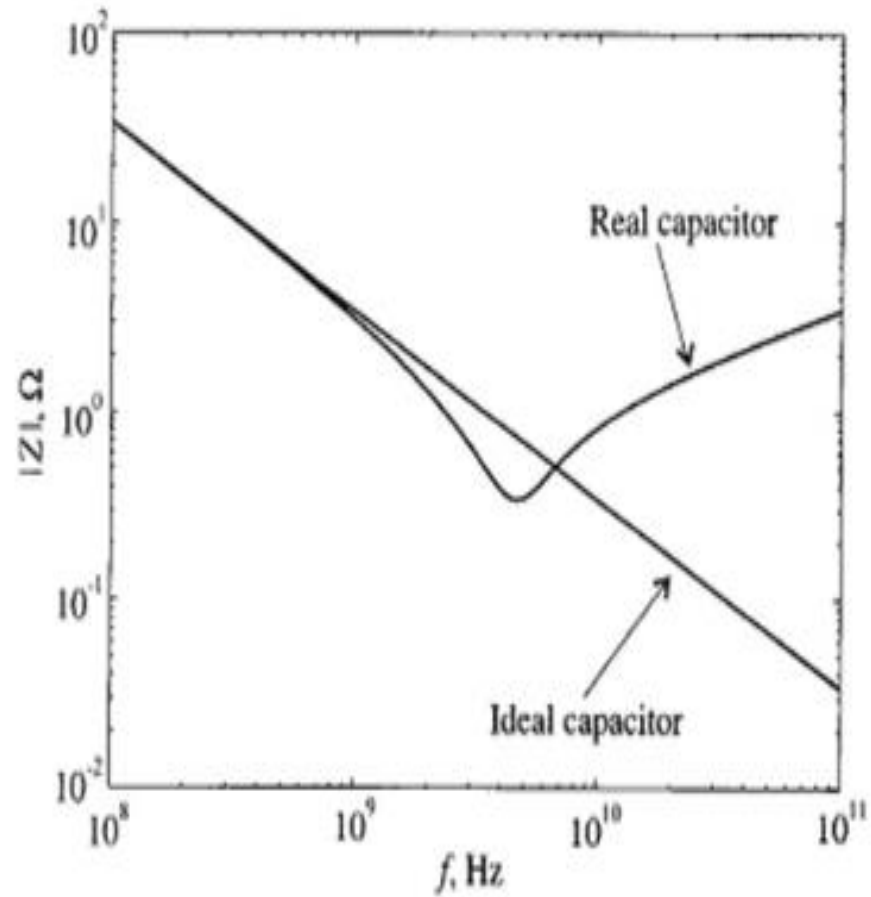
Ideal



$$Z(j\omega) = \frac{1}{j\omega C} = \frac{1}{\omega C} \angle -90^\circ$$

$$|Z| = \frac{1}{2\pi C} \cdot \frac{1}{f}$$





$$Z(j\omega) = R + j\omega L + \frac{1}{j\omega C}$$

$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

**At microwave frequencies
everything is combination of R, L
and C.**

Behavior of Circuits at Conventional and RF/Microwave Bands

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Low Frequency/ Conventional

- ▶ Bandwidths are limited hence small no. of channels can be adjusted.
- ▶ It uses lumped element circuit theory.
- ▶ Current flow and voltage drops are used to calculate power.
- ▶ Open wire, twisted cables, co-axial cables are used as transmission lines.

RF/Microwaves

- ▶ Provide large bandwidth so its possible to adjust large no. of channels.
- ▶ It uses distributed circuit theory (ohm/m, H/m, F/m).
- ▶ Scattering phenomena like absorption, reflection, refraction, etc. are used in power calculation.
- ▶ Optical fibers, waveguides, strip lines, micro-strip lines are common transmission lines.

Behavior of Circuits at Conventional and RF/Microwave Bands

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Low Frequency/ Conventional

- ▶ Lumped circuit elements are used such as resistors, filters, oscillators, etc.
- ▶ It uses current modulated mode.
- ▶ Almost all the solid state devices can be used.
- ▶ It can handle low power.

RF/Microwaves

- ▶ Cavity resonators or resonant lines are used as oscillators, resonators, etc.
- ▶ Density modulation or velocity modulation are used using magnetrons, klystrons, TWTs, etc.
- ▶ Vacuum tube like devices, micro-miniaturised solid state devices like Gunn diodes, tunnel diodes, IMPITT, TRAPPIT, etc. are used.
- ▶ It can handle higher power.

Assignment #1:

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1. Mention the microwave frequency band specified by IEEE. What are the advantages and disadvantages of using microwave signal?
2. Define microwave signal? Explain the characteristics behavior of passive component at microwave frequency.
3. Compare and contrast the circuit behavior in conventional low frequency and RF/Microwave frequency band. What are the application areas of microwave system?
4. Explain the general microwave system with clear diagram. Explain the characteristics of microwave signal.
5. Differentiate between microwave and acoustic/seismic waves.

► ***Due Date: 1 week from the slide presentation***



Thank You

ANY QUERIES???